

幾何学特論 IIB

講義：善本潔

2026 年度 後期

目次

第 1 章	講義	2
1.1	定義	2
第 2 章	演習問題	3

第 1 章

講義

1.1 定義

Def.1.1.0(グラフ : Graph)

2つの集合 $V(G)$ と $E(G)$ とその関係 φ によって定義される。

$$\varphi: E(G) \rightarrow \binom{V(G)}{2} \cup \binom{V(G)}{1}$$

第 2 章

演習問題

Example 1.1.37:

Structure of the Petersen graph Using $[5] = \{1, 2, 3, 4, 5\}$ as our 5-element set, we write the pair $\{a, b\}$ as ab or ba . Since 12 and 34 are disjoint, they are adjacent vertices when we form the graph, but 12 and 23 are not. For each 2-set ab , there are three ways to pick a 2-set from the remaining three elements of $[5]$, so every vertex has degree 3.

Thm 1.2.1:

Let $P(n)$ be a statement with an integer parameter n . If the following two conditions hold, then $P(n)$ is true for each positive integer n

- (i) $P(1)$ is true.
- (ii) For all $n > 1$, " $P(k)$ is true for $1 \leq k < n$ " implies " $P(n)$ is true."

Lem 1.2.5:

Every u, v - walk contains a u, v - path.

Prop 1.2.11:

Every graph with n vertices and k edges has at least $n - k$ components.

Thm 1.2.14:

An edge is a cut-edge if and only if it belongs to no cycle.

Lem 1.2.15:

Every closed odd walk contains an odd cycle.

Thm 1.2.18 (König):

A graph is bipartite if and only if it has no odd cycle.